Year 9 Mathematics

Unit 2: Area and Volume

2.3 Circumference of a Circle

Circumference of a Circle

STARTER · 5 MIN

Date: _____ **Title: Circumference of a Circle**

LEARNING OBJECTIVES

- Name the parts of a circle
- Calculate the circumference of a circle
- Solve problems with circumference

STARTER *Answer in your book. No calculators.*

1. The radius is 6 cm. What is the diameter?
2. Round 3.14159 to 1 d.p.
3. Round 3.14159 to 2 d.p.
4. Work out $2 \times 7 \times 3$

Starter — Answers

STARTER

Tick or correct in green pen

1. **12 cm**
2. **3.1**
3. **3.14**
4. **42**

Success Criteria

SUCCESS CRITERIA

By the end of this lesson:

BRONZE	I can name the parts of a circle	☐
SILVER	I can calculate the circumference of a circle	☐
GOLD	I can solve problems with circumference	☐

Tick each box in the plenary when you can do it on your own.

Introduction — Circumference of a Circle

INTRODUCTION · 3 MIN

The circumference is the distance all the way round a circle.

Parts

Radius r: centre to the edge.

Diameter d = 2r, all the way across.

Formula

Use the π button on your calculator.

$C = \pi d$ or $C = 2\pi r$

In terms of π

Leave π in the answer.

d = 12 cm: $C = 12\pi$ cm

Semicircle

Half the circumference plus the diameter.

$$P = \frac{1}{2}\pi d + d$$

Examples 1 and 2 — Finding the Circumference

EXAMPLES · 5 MIN

BRONZE

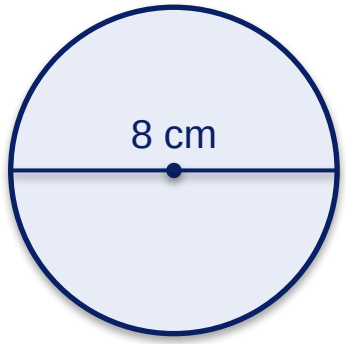
Watch and copy into your book

Hint $C = \pi d$ $C = 2\pi r$ $d = 2r$

Example 1

Find the circumference.

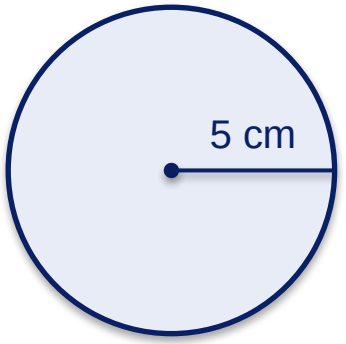
Give your answer to 1 d.p.



Example 2

Find the circumference.

Give your answer to 1 d.p.



Examples 1 and 2 — Corrections

CORRECTIONS

BRONZE

Check your copy and correct in green pen

Example 1

Diameter given: C = πd

C = π × 8

= 25.132…

= 25.1 cm (1 d.p.)

Tip: write the long answer before rounding.

Example 2

Radius given: C = 2πr

C = 2 × π × 5

= 31.415…

= 31.4 cm (1 d.p.)

Tip: or double the radius and use πd.

Task 1

PRACTICE · 9 MIN

BRONZE

Find the circumference. Give your answers to 1 d.p.

Hint $C = \pi d$ $C = 2\pi r$ $d = 2r$

1. diameter 10 cm
2. diameter 3 cm
3. radius 4 cm
4. radius 7.5 cm
5. A wheel has diameter 26 inches.

★ radius 0.6 m

Task 1 — Answers

CORRECTIONS

BRONZE

Tick or correct in green pen

1. **31.4 cm**
2. **9.4 cm**
3. **25.1 cm**
4. **47.1 cm**
5. **81.7 inches**

★ **3.8 m**

Examples 3 and 4 — In Terms of π and Semicircles

EXAMPLES · 5 MIN

SILVER

Watch and copy into your book

Hint C = πd semicircle: P = $\frac{1}{2}$πd + d

Example 3

Find the circumference in terms of π.

(a) d = 12 cm (b) r = 7 cm

Example 4

Find the perimeter.

Give your answer to 1 d.p.



10 cm

Examples 3 and 4 — Corrections

CORRECTIONS

SILVER

Check your copy and correct in green pen

Example 3

Leave π in the answer

(a) $C = \pi \times 12 = 12\pi$ cm

(b) $C = 2 \times \pi \times 7$

$= 14\pi$ cm

Tip: write the number before π.

Example 4

Curved part + straight edge

$\frac{1}{2} \times \pi \times 10 = 15.707\ldots$

$15.707\ldots + 10$

= 25.7 cm (1 d.p.)

Tip: do not forget the diameter.

Task 2

PRACTICE · 9 MIN

SILVER

Questions 1–2: in terms of π. 3–6: perimeter to 1 d.p.

Hint C = πd semicircle: P = $\frac{1}{2}$πd + d

1. Circumference: d = 9 cm
2. Circumference: r = 11 cm
3. Semicircle, diameter 8 cm
4. Semicircle, diameter 20 cm
5. Semicircle, radius 3 cm

★ Semicircle, diameter 14 m

Task 2 — Answers

CORRECTIONS

SILVER

Tick or correct in green pen

1. 9π cm
2. 22π cm
3. 20.6 cm
4. 51.4 cm
5. 15.4 cm

★ 36.0 m

Examples 5 and 6 — Circumference Problems

EXAMPLES · 5 MIN

GOLD

Exam-style: show all your working

Hint $d = C \div \pi$ $r = C \div 2\pi$
1 m = 100 cm, 1 km = 100 000 cm

Example 5

(2 marks)

The circumference of a circle is 50 cm. Find its diameter. Give your answer to 1 d.p.

Example 6

(4 marks)

A wheel has diameter 60 cm.

(a) How far does it go in 50 turns? Answer in metres.

(b) How many full turns does it make in 1 km?

Examples 5 and 6 — Corrections

CORRECTIONS

GOLD

Check your copy and correct in green pen

Example 5

C = πd, so d = C ÷ π

d = 50 ÷ π

= 15.915…

= 15.9 cm (1 d.p.)

Tip: radius = C ÷ 2π

Example 6

One turn = one circumference

C = 60π = 188.49… cm

(a) 50 × 188.49… cm = 94.2 m

(b) 100 000 ÷ 188.49 = 530.5

530 full turns

Tip: 1 km = 100 000 cm

Task 3

PRACTICE · 9 MIN

GOLD

Give your answers to 1 d.p. unless told otherwise

Hint $d = C \div \pi$ $r = C \div 2\pi$
1 m = 100 cm, 1 km = 100 000 cm

1. C = 40 cm. Find d.

2. C = 100 m. Find d.

3. C = 25 cm. Find r.

4. Wheel: d = 70 cm.
Full turns in 1 km?

5. Wheel: r = 30 cm.
Distance in 100 turns, in m?

★ Circular track: C = 400 m.
Find r.

Task 3 — Answers

CORRECTIONS

GOLD

Tick or correct in green pen

1. **12.7 cm**
2. **31.8 m**
3. **4.0 cm**
4. **C = 219.9… cm, 454 full turns**
5. **188.5 m**

★ **63.7 m**

Plenary — Exit Ticket

PLENARY · 5 MIN

Answer in your book. Try all three.

BRONZE Find the circumference: d = 6 cm (1 d.p.)

SILVER Find the circumference in terms of π: r = 5 cm

GOLD C = 20 cm. Find the diameter (1 d.p.).

Reflect: Look back at the learning objectives. Which ones can you do now?

Plenary — Answers

PLENARY

Tick or correct in green pen

BRONZE **18.8 cm**

SILVER **10π cm**

GOLD **6.4 cm**

Exam Practice — Circumference

EXAM PRACTICE · 10 MIN

BRONZE

IGCSE exam-style: show all your working

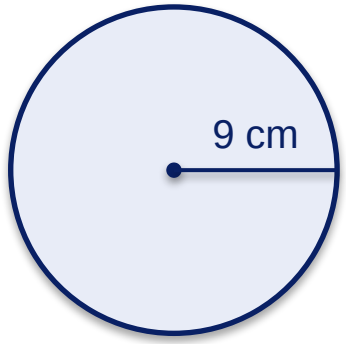
Hint C = πd C = 2πr d = 2r

Question 1

(2 marks)

Find the circumference.
Give your answer to 1 d.p.

Diagram NOT accurately drawn



..... cm

Question 2

(2 marks)

The wheels on Aysha's bicycle have a diameter of 28 inches.
Work out the circumference of one wheel to 1 d.p.

..... inches

Exam Practice — π and Semicircles

EXAM PRACTICE · 10 MIN

SILVER

IGCSE exam-style: show all your working

Hint C = πd semicircle: P = $\frac{1}{2}$πd + d

Question 3

(2 marks)

Find the circumference in terms of π.

(a) diameter 15 cm (1)

..... cm

(b) radius 4 cm (1)

..... cm

Question 4

(2 marks)

Find the perimeter of this semicircle to 1 d.p.

Diagram NOT accurately drawn



..... cm

Exam Practice — Circumference Problems

EXAM PRACTICE · 10 MIN

GOLD

IGCSE exam-style: show all your working

Hint
$d = C \div \pi$ $r = C \div 2\pi$
1 m = 100 cm, 1 km = 100 000 cm

Question 5

(2 marks)

The circumference of a circle is 66 cm.
Find its radius to 1 d.p.

..... cm

Question 6

(3 marks)

A wheel has diameter 65 cm.
It turns 200 times.
How far does it travel?
Give your answer in metres to 1 d.p.

..... m

Exam Practice — Mark Scheme

CORRECTIONS

BRONZE

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 1

$2 \times \pi \times 9$ — M1

= 56.5 cm — A1

Tip: radius given: C = 2πr

Question 2

$\pi \times 28$ — M1

= 88.0 inches — A1

Tip: diameter given: C = πd

Exam Practice — Mark Scheme

CORRECTIONS

SILVER

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 3

(a) 15π cm B1

(b) 8π cm B1

Tip: 2 × 4 = 8, so 8π

Question 4

$\frac{1}{2} \times \pi \times 14 = 21.99\ldots$ M1

+ 14 = 36.0 cm A1

Tip: add the straight edge.

Exam Practice — Mark Scheme

CORRECTIONS

GOLD

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 5

$r = 66 \div (2 \times \pi)$ — M1

= 10.5 cm — A1

Tip: $C = 2\pi r$, so $r = C \div 2\pi$

Question 6

$C = 65\pi = 204.2\ldots$ cm — M1

$\times 200 = 40\,840.7\ldots$ cm — M1

= 408.4 m — A1

Tip: 100 cm = 1 m